NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Osteopathic Considerations in Pulmonary Disease – 2

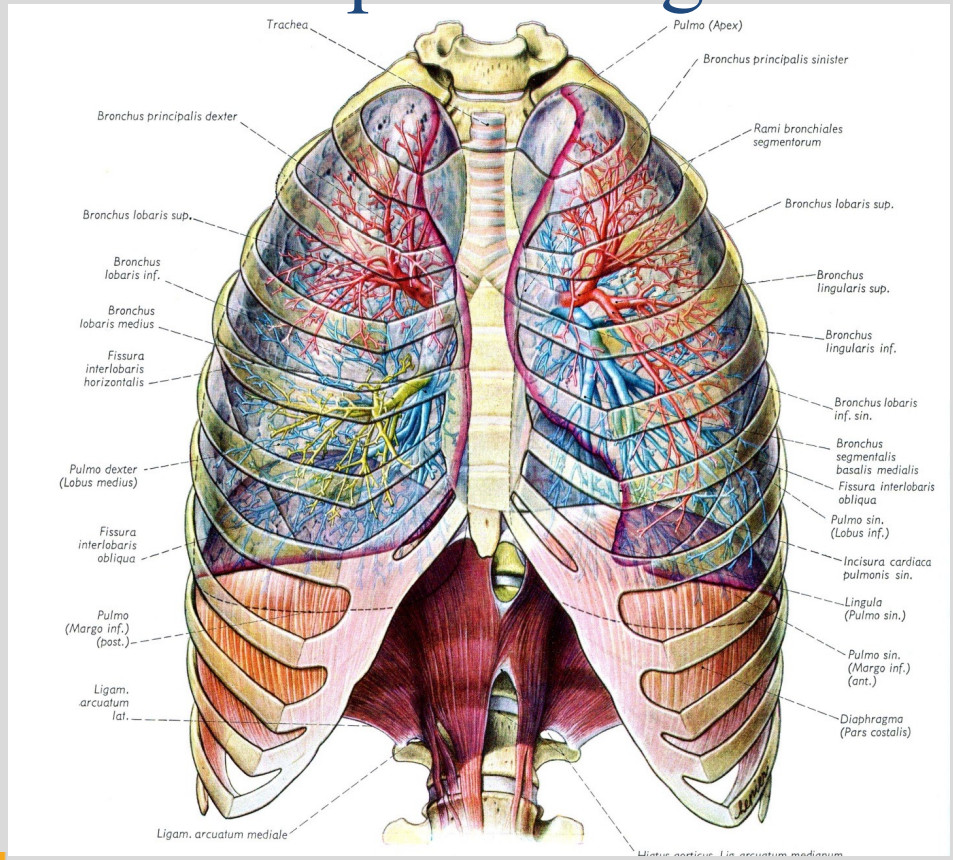
Hugh Ettlinger, DO, FAAO, FCA
OMM
Hughettlinger@Hotmail.com

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1) Understand the role of mechanics, including work of breathing, thoracic cage compliance, and respiratory muscle fatigue in COPD and ventilator dependence.
2) Understand the design of the pulmonary lymphatics
3) Understand the role of the lymphatic system in pneumonia and asthma, and how to examine and treat in these diseased.
4) Understand the physiologic basis for the treatment of asthma, and how to use the osteopathic structural exam to individualize treatment of the asthmatic patient

Forces producing air exchange



Source: Pernkopf atlas of anatomy

- The negative intrathoracic pressure which produces air exchange is created by the action of skeletal muscles on the osseous thorax. The lung parenchyma is passive

COPD

- Compliance – Severely reduced (thoracic cage)
  - Work of breathing
  - Functional capacity
- Respiratory muscle fatigue
  - Acute – respiratory failure
  - Chronic – identified in COPD patients

VENTILATOR

- After a few days, it can become increasingly difficult to wean a patient from the ventilator
  - Fluid
  - Mechanics

VENTILATOR

- After a few days, it can become increasingly difficult to wean a patient from the ventilator
  - Fluid

VENTILATOR

- Pulmonary Drainage
  - There is 70% less pulmonary lymph drainage in mechanically ventilated animals compared to spontaneous breathing
  - This applies to pleural as well as pulmonary lymph
  - Can be seen throughout the body - anasarca

VENTILATOR

1. Frostell, C., H. Blomqvist, G. Hedenstierna, I. Halbig, and R. Pieper. "Thoracic and Abdominal Lymph Drainage in Relation to Mechanical Ventilation and PEEP." *Acta Anaesthesiologica Scandinavica* 31.5 (1987): 405-12. Web.

2. Moriondo, Andrea, Sylvain Mukenge, and Daniela Negrini. "Transmural Pressure in Rat Initial Subpleural Lymphatics during Spontaneous or Mechanical Ventilation." *AJP: Heart and Circulatory Physiology* 289.1 (2005): 263-69. Web.

VENTILATOR

- After a few days, it can become increasingly difficult to wean a patient from the ventilator
  - Mechanics

VENTILATOR

- Mechanics
  - Compliance changes in the osseous thorax are obvious on Osteopathic exam
  - Reduced drainage will increase surface tension, predisposing to atelectasis and increasing airway resistance, along with the increased secretions in the larger airways associated with the ET tube

VENTILATOR

- Mechanics
  - Respiratory muscles not only fatigue, but decompensate and atrophy, including the diaphragm
  - Diaphragm atrophy has been demonstrated at 18 hours
  - The longer the patient is ventilated, the worse the issue becomes, and can reduce size up to 57%

VENTILATOR

1. Tobin, Martin J. "Narrative Review: Ventilator-Induced Respiratory Muscle Weakness." *Annals of Internal Medicine* 153.4 (2010): 240-45.

2. Levine, Sanford. "Rapid Disuse Atrophy of Diaphragm Fibers in Mechanically Ventilated Humans." *The New England Journal of Medicine* 358.13 (2008): 1327-335.

3. Hooijman, Pleuni E. "Diaphragm Muscle Fiber Weakness and Ubiquitin–Proteasome Activation in Critically Ill Patients." *American Journal of Respiratory and Critical Care Medicine* 191.10 (2015): 1126-138.

VENTILATOR

- Approach
  - Improve compliance and excursion of thorax
  - Improve respiratory muscle function
  - Lymphatic drainage will not improve significantly until the patient is weaned
  - Encouragement – *“failed weaning trials”*

PNEUMONIA

- Long hx of successful clinical success
  - 1918 flu pandemic
  - Retrospective study
- National fatality rates
  - 6% flu
  - 33% flu complicated with pneumonia
- Patients treated in Osteopathic hospitals receiving OMT
  - 110,000 cases flu, 0.25% fatality
  - 6250 cases flu + pneumonia, 10% fatality

Pneumonia

- Tuttle, LK “Influenza and Pneumonia Treatment”
- The Journal of the American Osteopathic Association September 1919 | Vol 18 | 211-214

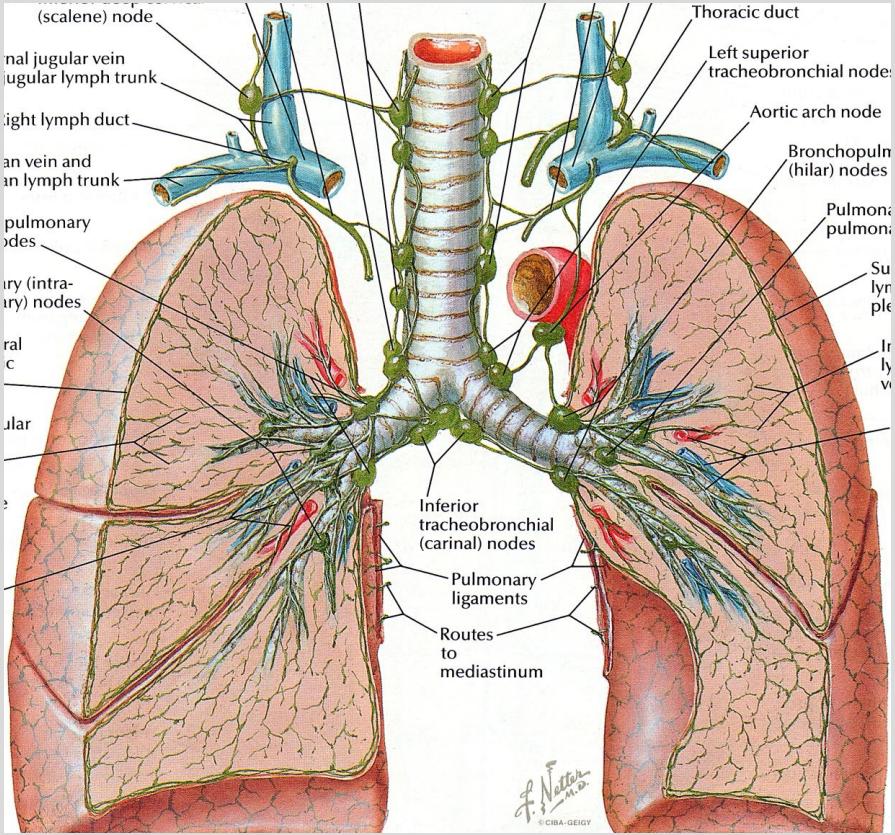
Pneumonia

•Noll, Donald R.,, et. al “Multicenter Osteopathic Pneumonia Study in the Elderly: Subgroup Analysis on Hospital Length of Stay, Ventilator-Dependent Respiratory Failure Rate, and In-hospital Mortality RateThe Journal of the American Osteopathic Association September 2016 | Vol 116 | No. 9

- OMT group had a shorter LOS than the CCO group (median, 3.8 days [n=40] vs 5.0 days [n=50]; P=.01
- lower ventilator-dependent respiratory failure rate than the CCO group (0 of 40 [0%] vs 5 of 50 [10%]; P=.05)
- in-hospital mortality rates in the PSI class V subgroup were lower for the OMT group (1 of 22 [5%]) than the CCO group (6 of 19 [32%]; P=.05)

PNEUMONIA

- Lymphatic system
  - Pneumonia produces significant inflammation of the small airways, with significant secretions throughout

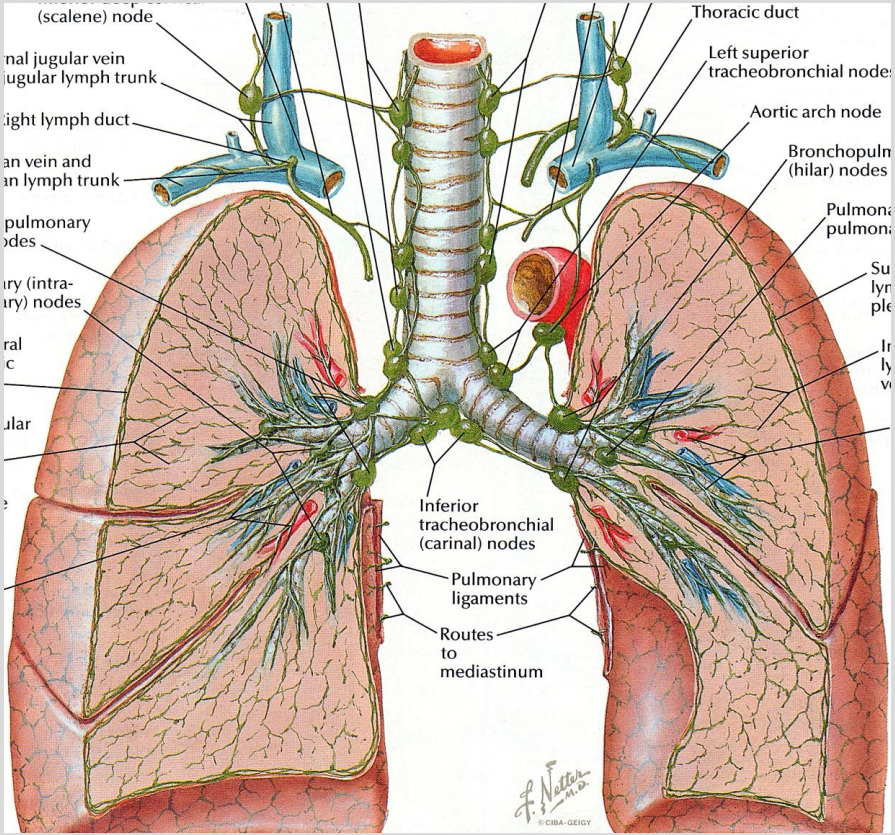


Source: Netter, Atlas of Human Anatomy

Pulmonary lymphatics follow not only the vasculature, but also the airways.

The airways are drained by the lymphatica as well as the mucociliary mechanism.

The mucociliary mechanism does not extend to the terminal bronchioles, or the alveoli. The entire drainage of these terminal airways occurs via the lymph.

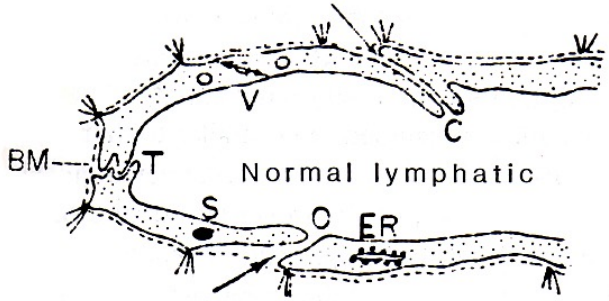


Source: Netter, Atlas of Human Anatomy

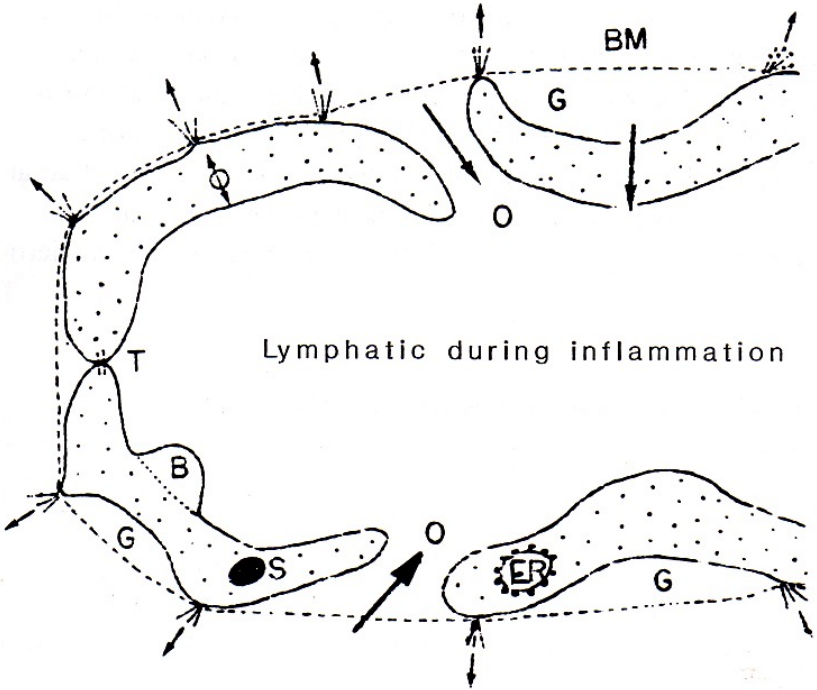
Pneumonia involves significant inflammation of the small airways, which are drained through the lymphatics.

Pulmonary lymph is pumped by respiratory excursion, which alternately opens and closes the initial lymphatics.

Respiratory excursion is significantly reduced in pneumonia



EXHALATION



INHALATION

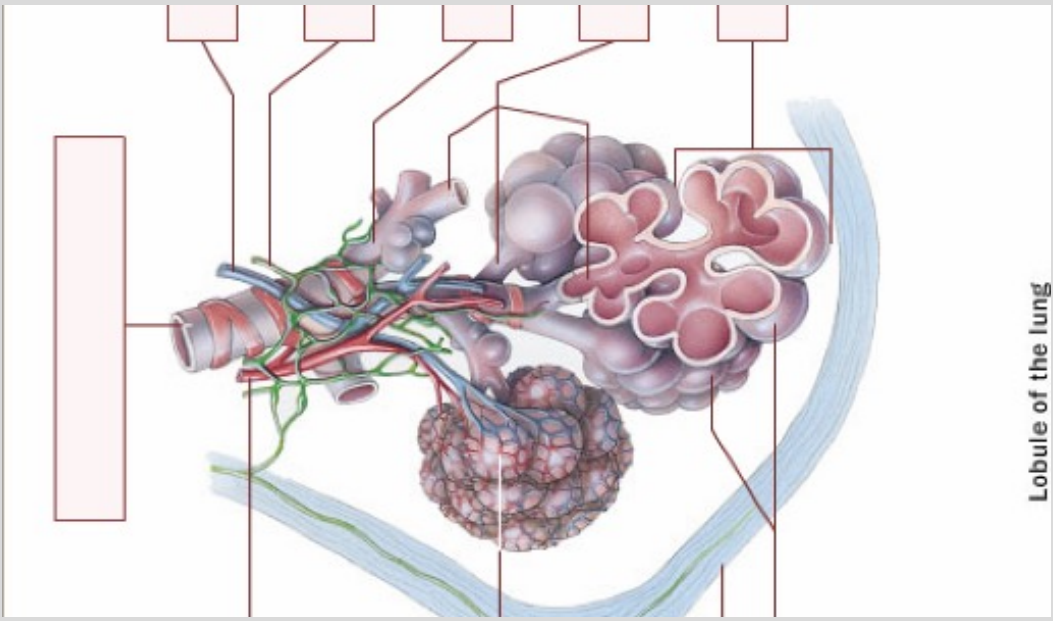
PNEUMONIA

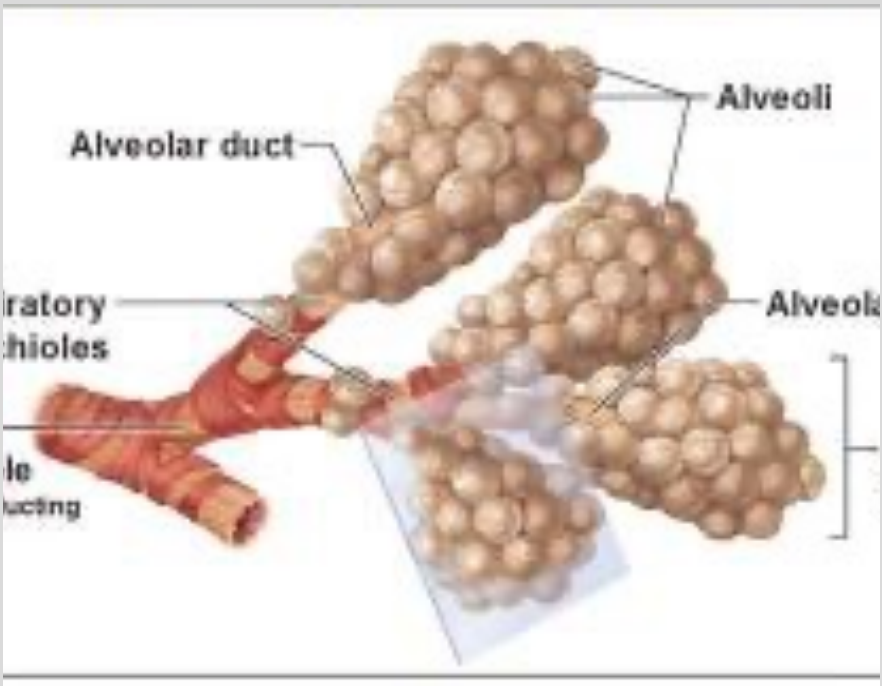
- “Diminished local excursion of the Thorax may be caused by... Consolidation”

DeGowin and DeGowin, Bedside Diagnostic Exam.

PNEUMONIA

- Lymphatic system
  - Aveolar lymphatics have been discovered and are thought to be “the absorbing compartment of the lung responsible for maintaining the alveolar interstitium relatively dry in order to provide a minimal thickness of the air-blood barrier and thus optimize gas diffusion.


Lobule of the lung


Alveoli
Alveolar duct
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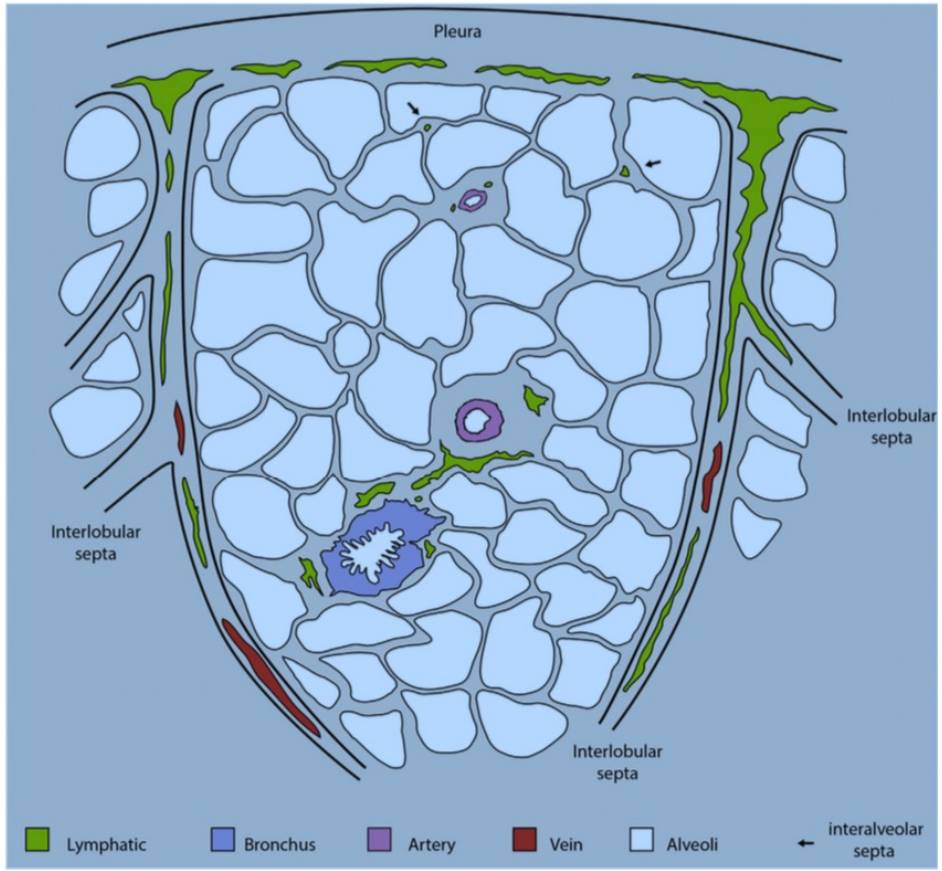


Fig. 3. Distribution of lymphatic vessels (green) in the pulmonary lobule. Most vessels are located in the pleura, in the interlobular septa and in association with bronchovascular bundles. Lymphatic vessels are also present in interalveolar septa, in association with arterioles and only occasionally independent of blood vessels. Arrows indicate lymphatic vessels independent of blood vessels in interalveolar septa.

PNEUMONIA

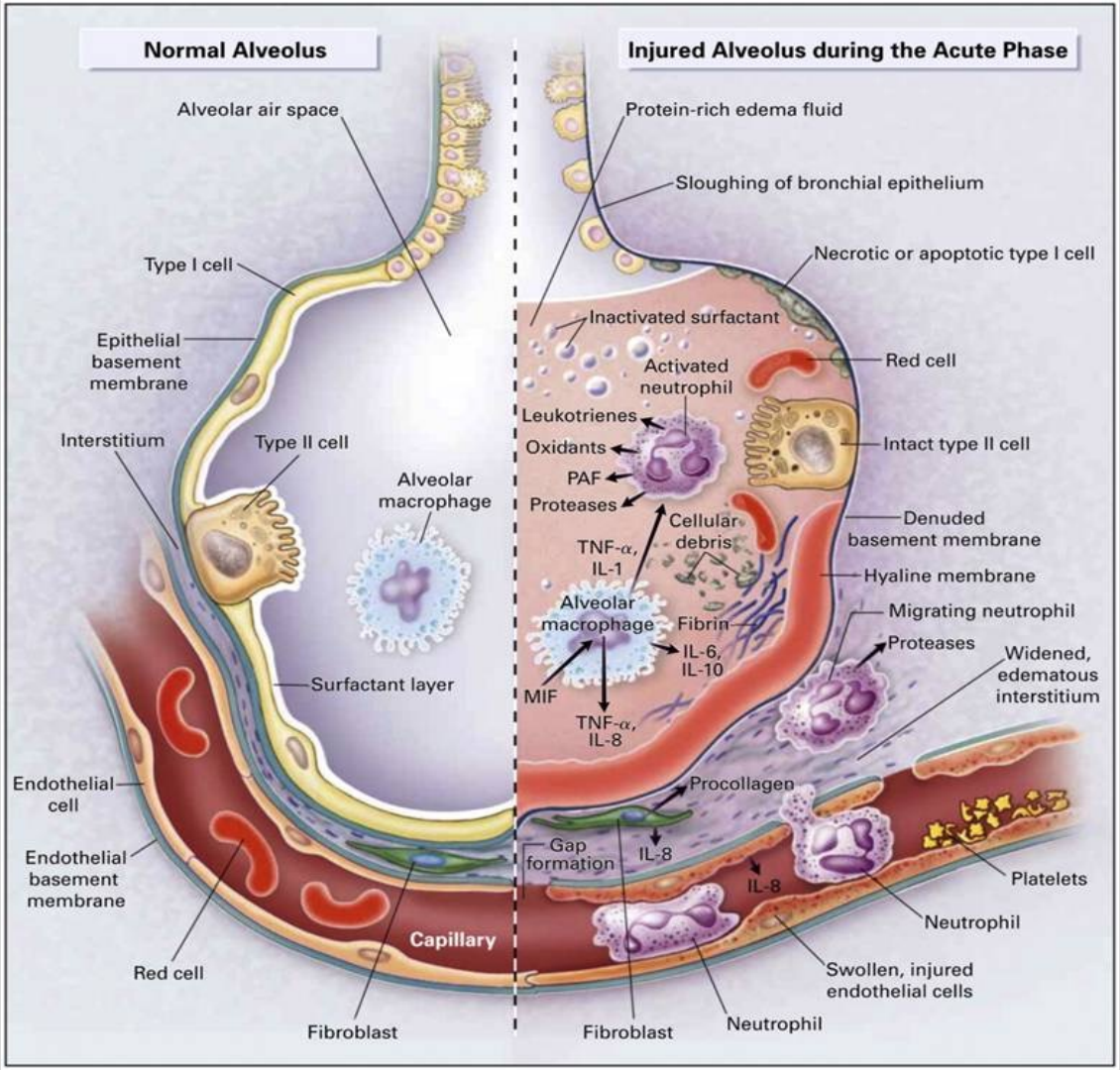
- Approach
  - Improve excursion of thorax
  - Improve diaphragm function
  - Lymphatic drainage procedures

PNEUMONIA

- COVID
  - Extreme inflammation
  - ARDS
  - Fibrosis
  - *Cytokine Storm*

PNEUMONIA

- ARDS
  - Interaction between alveolus and capillary endothelium
    - Diffuse alveolar wall damage
    - Fibrin deposits and fibrosis


Normal Alveolus
Injured Alveolus during the Acute Phase
Alveolar air space
Type I cell
Epithelial basement membrane
Interstitium
Type II cell
Alveolar macrophage
Surfactant layer
Endothelial cell
Endothelial basement membrane
Red cell
Capillary
Fibroblast
Protein-rich edema fluid
Sloughing of bronchial epithelium
Necrotic or apoptotic type I cell
Inactivated surfactant
Activated neutrophil
Red cell
Leukotrienes
Oxidants
PAF
Proteases
Intact type II cell
Cellular debris
TNF-α, IL-1
Denuded basement membrane
Hyaline membrane
Alveolar macrophage
Fibrin
Migrating neutrophil
Proteases
IL-6, IL-10
MIF
TNF-α, IL-8
Widened, edematous interstitium
Procollagen
Gap formation
IL-8
IL-8
Platelets
Neutrophil
Swollen, injured endothelial cells
Fibroblast
Neutrophil

PNEUMONIA

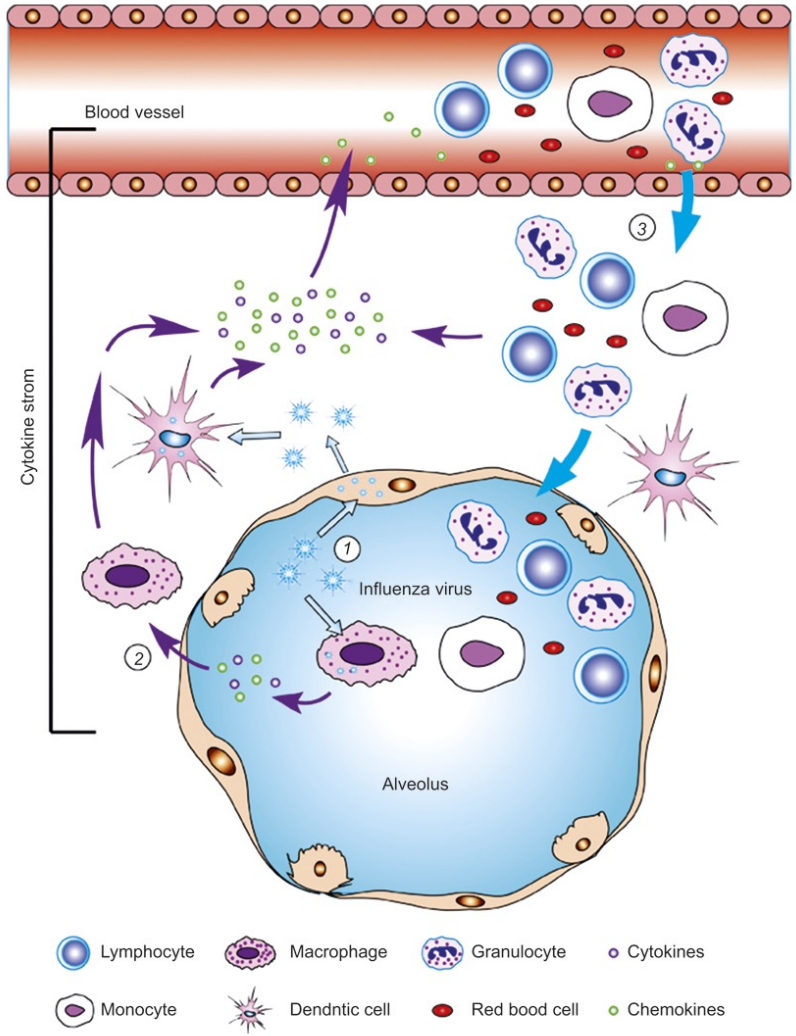
- ARDS
  - “All these data suggest that the wide-spread lung damage associated with SARS may be caused more by an exaggerated immune response than the virus itself.”
  - Mice that did not develop lymphatic in utero essentially were born in ARDS

PNEUMONIA

- Fibrosis
  - Role of Lymph appears similar to other organs
  - “Two models of lung fibrosis suggest that impairment of lung lymphatics could be key to the development of fibrosis”

PNEUMONIA

- Cytokine storm
  - Systemic cytokine storm – multiorgan failure
  - "Moreover, the severe inflammatory cytokines/chemokines can *spill over* into the circulation and result in systemic cytokine storms, which are responsible for multi-organ dysfunction."
  - Lymph?


Blood vessel
Cytokine strom
1
Influenza virus
2
3
Alveolus
Lymphocyte
Macrophage
Granulocyte
Cytokines
Monocyte
Dendntic cell
Red bood cell
Chemokines

PNEUMONIA

- COVID
  - SBH
  - *Intubated patients – Over 80% death rate – consistent with NYS and national numbers*
  - *OMT 14 treated – 8 recovered and were discharged*

ASTHMA

- Asthma is a complex clinical condition

ASTHMA

- Effects of Osteopathic Manipulative Treatment on Pediatric Patients With Asthma: A Randomized Controlled Trial
- JAOA • Vol 105 • No 1 • January 2005 • 7-12
- PEF increased 7L/m to 9L/m vs control. P=0.95 140 patients

ASTHMA

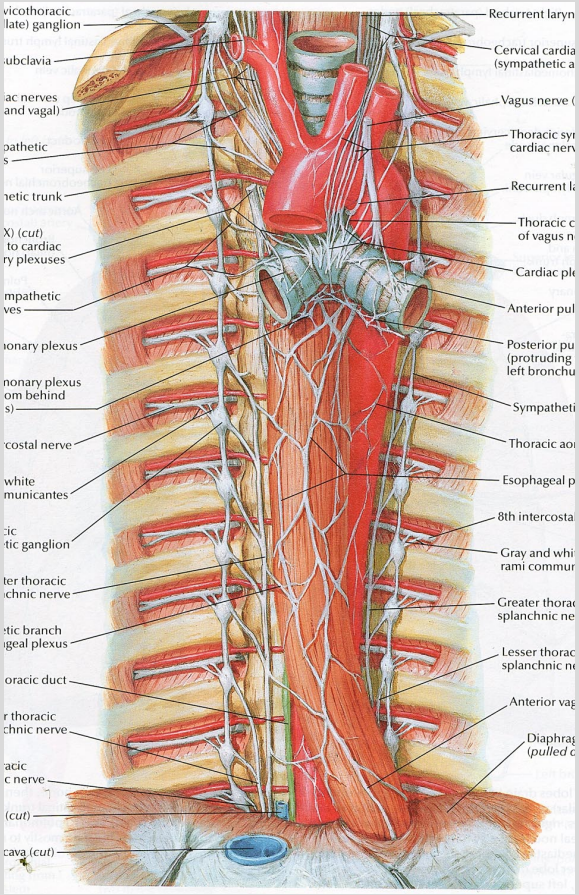
- Asthma is a complex clinical condition
- Emergency situations
  - Asthmatics go into respiratory failure due to muscle fatigue/exhaustion
  - Same concepts of WOB, mechanics apply
    - Tight vs Stiff (reversible?)

ASTHMA

- Asthma is a complex clinical condition
- Reactive airway
  - Beta agonists
  - Reflex facilitation

Innervation

- There is both a sympathetic and parasympathetic innervation of the lungs.

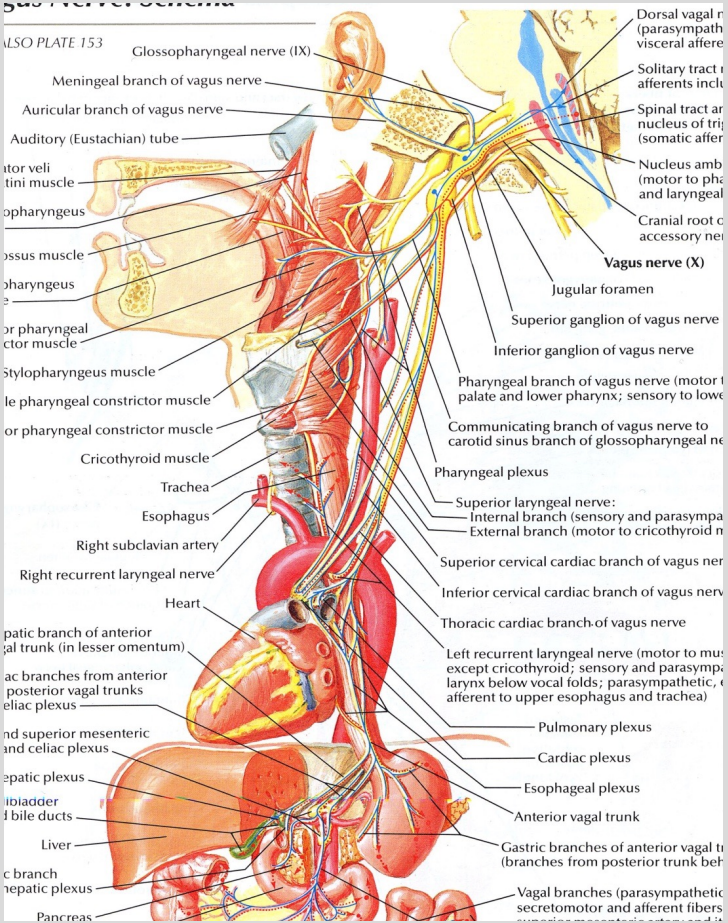


Netter, Atlas of Human Anaomy

The sympathetics to the lungs have origin from thoracic segments T1-T5. The sympathetic innervation of the lungs causes bronchodilation, and reduces mucous secretion, although it may increase the viscosity of the secretions.

There is clinical evidence that the sensory fibers from the lungs (nociceptors) travel predominantly with the vagus. The nociceptors which travel with the sympathetics arise mostly from the large airways.

Upper thoracic reflex dysfunction is most commonly found in Bronchitis.



Netter, Atlas of Human Anatomy

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The vagus is the parasympathetic innervation of the lungs. It produces bronchoconstriction and increases mucous production.

Most nociceptors from the smaller airways are carried by the vagus, so asthma and other small airway diseases will produce reflexes via this pathway.

The vagus has been shown to reflex with the upper cervical spine, most commonly at C2. The sensory ganglia of the vagus are found at the OA and jugular foramen.

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ASTHMA

- Asthma is a complex clinical condition
- Inflammatory Disease
  - Steroids
  - Lymphatic Drainage
  - Respiratory excursion is significantly reduced in Asthma

ASTHMA

- Asthma is a Neuroendocrine disease
- Reactive airway
  - Stimulation of Pulmonary Nociceptors causes them to dispense pro-inflammatory peptides
- Inflammatory Disease
  - Inflammation sensitizes nociceptors lowering their threshold for firing

ASTHMA

- Approach
  - Improve compliance and excursion of thorax – reduce work of breathing
  - Improve lymphatic drainage
  - Reduce segmental facilitation

ASTHMA

- Approach
  - Can Osteopathic structural exam distinguish the asthmatic with a predominantly reactive airway from one who is predominantly inflamed?
    - Cranio-cervical dysfunction
    - Tight/congested chest
    - Beta-adrenergic vs. steroids

Resources / Reading Assignment: 10 pages

- Osteopathic Approach to Diagnosis and Treatment DSD Chp 112: Pulmonary Applications

- Foundations of Osteopathic Medicine (FOM), 3rd Ed., Chapter 13 "Mechanics of Respiration," especially: Read: **"Introduction" ; "The Pumphead of the Thoracic Cylinder- Thoracic Diaphragm"; "Airway Diseases- Structural Changes in COPD"; "Kyphosis"**

- Foundations of Osteopathic Medicine (FOM), 3rd Ed., Chp 39 "Thoracic Region and Rib Cage," especially: "Introduction" upto "Musculature"; and Figure 39.3 to "Lymphatics"; "Neural Structures" ; "Biomechanical Considerations" to the End

Summary Slide

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- There is an Osteopathic Approach to help patients with all pulmonary diseases
- Breathing is a mechanical process, and is important is respiratory diseases, especially Asthma/COPD, and with managing patients on Ventilators
- Lymph drainage of the lungs is driven by respiratory excursion. Improving it will help with pneumonia, Asthma, and other conditions with significant inflammation
- Reducing facilitation may reduce the reactivity of the airway in Asthma. The upper cervical spine and cranio-cervical junction is the most important place to examine and treat

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